

Controller-Shaped Cycle-Space Escape Geometry for Nonlinear Stability of Inverter-Dominated Power Networks

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Abstract—Small-signal eigenvalues, graph-connectivity measures, and stochastic linearized escape metrics answer different stability questions and can rank the same network differently. This paper develops a cycle-space framework aimed at the missing question: *what is the smallest physically structured perturbation that can drive a specific inverter-dominated grid out of a safe nonlinear operating region, through which network structures does the escape propagate, and where should control be modified to block it?* The starting point is an exact cycle-flow representation of lossless AC equilibria. Within each winding sector, a convex cycle potential has Hessian $C^\top K_e^{-1} C$, where C spans the graph cycle space and K_e is the operating-point tangent synchronizing stiffness. This yields a basis-invariant *cycle-compliance spectrum* through a generalized eigenproblem and a closed-form local cycle-flow boundary radius. Controller dynamics are then introduced as frequency-dependent matrix transports. Their cycle holonomies separate naturally into an Abelian component, captured by the determinant, and a genuinely non-Abelian component measured by noncommuting special-linear factors. The transient extension is formulated as a structured minimum-action escape problem on the nonlinear DAE, allowing disturbances to be constrained to buses, lines, controllers, cycles, or zones. Linear reachability provides a closed-form local baseline; in the conservative limit, barrier sensitivities follow directly from stable and saddle flows, while the general controller-aware problem is posed through an adjoint optimal-control system. A cycle-weighted projection of finite-time Lyapunov directions is proposed to test whether chaotic basin-boundary stretching is supported by the same structures that dominate small-signal feedback cycles. Preliminary AC-consistent IEEE 39-bus calculations reveal a concrete geometric reversal: adding line 25–38 increases generalized-Fiedler connectivity by 0.146% and standard Fiedler connectivity by 0.298%, yet the maximum cycle compliance increases by 6.42% and a local cycle-flow boundary radius decreases by 6.18%. The softest cycle direction, originally concentrated on the 26–28–29 triangle, spreads toward the new 25–38 route. These results do not yet establish a transient-stability reversal; rather, they identify a falsifiable candidate zone for the nonlinear minimum-action experiment. The proposed framework therefore separates four objects that should not be conflated: small-signal margin, scalar-model fidelity, cycle-geometric fragility, and nonlinear transient escape action.

Index Terms—Transient stability, inverter-dominated power systems, cycle space, winding numbers, nonlinear dynamics, minimum action, basin of attraction, graph theory, holonomy, non-Abelian feedback, chaos, virtual inertia, controller placement.

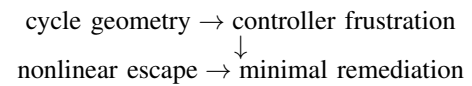
Research manuscript draft, August 2026. The cycle-space geometry and the IEEE 39-bus proof-of-concept are evaluated in this manuscript. The full nonlinear minimum-action, chaotic-boundary, and EMT/full-order validation stages are stated explicitly as remaining falsification gates rather than reported as completed experiments.

I. INTRODUCTION

POWER-SYSTEM stability is increasingly governed by a mixture of network physics and heterogeneous closed-loop converter dynamics. A bus that appears strong under short-circuit or Laplacian metrics can still participate in a poorly damped MIMO mode; conversely, a line reinforcement that improves connectivity can alter controller-network interactions in a way that reduces a closed-loop margin. Our companion work on gauge-invariant scalarizability addresses when a matrix-valued grid can be represented safely by decoupled scalar channels. The present paper asks a different and more nonlinear question: *how far is a specific operating point from a physically meaningful transient escape, where is the weakest entry point, which graph structures support the escape, and what is the least costly intervention that blocks it?*

Several ingredients are known separately. Cycle flows and winding numbers provide a rigorous description of phase-locked states and multistability in oscillator and swing-equation networks [1]. Convex formulations now allow systematic computation of synchronized states in cohesive regions [2]. Cycle space also enters stochastic synchronization robustness and line-vulnerability formulas [3]. Fluctuation-driven desynchronization and weak escape links have been studied analytically [4]; transient chaos and chaotic saddles have been observed in power-grid models with strongly nonuniform spatial vulnerability [5]. More recently, CREP introduced a parameter-design metric based on the invariant distribution of a *linearized* stochastic process [6], while domain-of-attraction boundaries have been approached through index-1 saddles, unstable periodic orbits, and stable manifolds [7]. These results make clear that another generic “stability index” would add little.

The target here is instead a structured bridge between four levels:



The distinguishing feature is localization. Disturbances and interventions are not allowed to act arbitrarily in state space: they are attached to physical buses, lines, devices, feedback cycles, or zones. This creates a direct engineering output: the location where the easiest unsafe perturbation enters, the network structures along which it develops, and the distinct

location where a small controller change most efficiently raises the escape barrier.

The contributions of this research draft are:

- C1.** A cycle-flow formulation of the nonlinear AC equilibrium equipped with a strictly convex cycle potential in the cohesive region. Its Hessian induces a *cycle-compliance geometry*.
- C2.** A basis-invariant cycle-compliance spectrum and a closed-form local cycle-flow feasibility radius. These quantities are geometric screening objects, not transient-stability claims.
- C3.** A controller-frustration decomposition in which scalar $U(1)$ -type phase frustration is the Abelian limit of matrix-valued cycle holonomy, while noncommuting special-linear factors capture irreducibly MIMO feedback frustration.
- C4.** A structured nonlinear minimum-action escape formulation for buses, edges, controllers, cycles, and zones, together with its linear Gramian baseline and conservative barrier sensitivities.
- C5.** A cycle-resolved test for whether finite-time chaotic stretching, basin-boundary escape, and controller feedback are supported by the same network structures.
- C6.** Preliminary IEEE 39-bus evidence that nodal connectivity and cycle-space compliance can move in opposite directions under the same AC-re-solved network upgrade, providing a concrete candidate for the subsequent nonlinear escape experiment.

II. POSITIONING: WHAT IS KNOWN AND WHAT IS NEW HERE

Table I separates established mathematics from the proposed synthesis. This distinction is important because winding numbers, basin stability, minimum-action methods, and holonomy are not new individually.

III. NONLINEAR CYCLE-SPACE FORMULATION

A. Cycle coordinates

Let $G = (\mathcal{V}, \mathcal{E})$ be a connected oriented network with incidence matrix $B \in \mathbb{R}^{n \times m}$. Let $C \in \mathbb{R}^{m \times \beta}$ have full column rank and span $\ker B$, where $\beta = m - n + 1$. For nodal injections p satisfying $\mathbf{1}^\top p = 0$, choose one feasible particular edge flow φ_0 with $B\varphi_0 = p$. Every other flow satisfying Kirchhoff's current law is

$$\varphi(q) = \varphi_0 + Cq, \quad q \in \mathbb{R}^\beta. \quad (1)$$

For the lossless sinusoidal relation

$$\varphi_e = a_e \sin(\delta_e - \alpha_e), \quad (2)$$

where $a_e > 0$ and α_e is an effective phase shift, the cohesive branch $|\delta_e - \alpha_e| < \pi/2$ gives

$$\delta(q) = \alpha + \arcsin[A^{-1}(\varphi_0 + Cq)]. \quad (3)$$

Phase consistency requires the cycle sums

$$F_C(q, n) = C^\top \delta(q) - 2\pi n = 0, \quad n \in \mathbb{Z}^\beta, \quad (4)$$

which partitions synchronized states into winding sectors [1].

B. Cycle potential and compliance geometry

Define, for $|f| < a$,

$$\phi_e(f) = f \arcsin(f/a_e) + \sqrt{a_e^2 - f^2}. \quad (5)$$

Then $\phi'_e(f) = \arcsin(f/a_e)$. For a winding sector n , consider

$$\Psi_n(q) = \sum_e \phi_e(\varphi_e(q)) + \alpha^\top \varphi(q) - 2\pi n^\top q. \quad (6)$$

Theorem 1 (Cycle-space Hessian). *Inside the cohesive region $|\varphi_e| < a_e$, the stationarity condition $\nabla_q \Psi_n = 0$ is equivalent to (4), and*

$$\nabla_q^2 \Psi_n(q) = C^\top K_e^{-1}(q) C, \quad (7)$$

where

$$K_e(q) = \text{diag} \left(\sqrt{a_e^2 - \varphi_e(q)^2} \right) = \text{diag} (a_e \cos(\delta_e - \alpha_e)). \quad (8)$$

Hence Ψ_n is strictly convex whenever C has full column rank and $K_e \succ 0$.

Proof. Differentiating (6) and using $\phi'_e(f) = \arcsin(f/a_e)$ gives (4). A second derivative gives $\phi''_e(f) = (a_e^2 - f^2)^{-1/2} = k_e^{-1}$. Therefore the Hessian is $C^\top \text{diag}(k_e^{-1}) C$, which is positive definite on the full-rank cycle coordinates when all $k_e > 0$. $\square$

Equation (7) gives a precise meaning to a *soft cycle*: it is a circulation direction that uses edges with large tangent compliance k_e^{-1} . As an edge approaches its sinusoidal limit, $k_e \rightarrow 0$ and the cycle metric becomes strongly distorted.

C. Basis-invariant cycle-compliance spectrum

The numerical matrix $C^\top K_e^{-1} C$ changes under a basis transformation $C \mapsto CR$. To remove this arbitrariness, define

$$C^\top K_e^{-1} C v = \chi C^\top C v. \quad (9)$$

Proposition 1 (Basis invariance). *The generalized eigenvalues $\{\chi_r\}_{r=1}^\beta$ of (9) are unchanged under any nonsingular cycle-basis transformation $C \mapsto CR$.*

Proof. Both matrices undergo the same congruence: $(G, M) \mapsto (R^\top G R, R^\top M R)$. Generalized eigenvalues are invariant under this simultaneous nonsingular congruence. $\square$

Equivalently,

$$\chi_{\max} = \max_{0 \neq z \in \ker B} \frac{z^\top K_e^{-1} z}{z^\top z}. \quad (10)$$

We call $\{\chi_r\}$ the *cycle-compliance spectrum*. It is not a stability spectrum: it measures operating-point-dependent softness of divergence-free edge-flow perturbations.

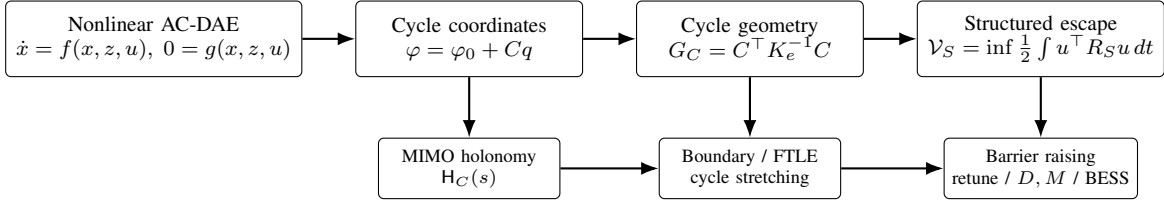


Fig. 1. Proposed theory. Cycle geometry provides an operating-point-dependent low-dimensional description of circulation softness; matrix-valued controller holonomies add dynamic frustration; minimum-action escape measures the perturbation needed to leave the safe region; and the dual optimization raises that barrier with the smallest intervention.

TABLE I
POSITIONING OF THE FRAMEWORK

Component	Established foundation	Role pursued here
Cycle flows / winding sectors	Stable phase-locked states can be parameterized using cycle flows and winding numbers [1], [2].	Use the cycle space as the coordinate system in which geometric softness and structured escape support are evaluated.
Stochastic robustness	Linearized stochastic phase-difference variance and CREP quantify fluctuation/escape risk [3], [6].	Replace unrestricted/global stochastic summaries by nonlinear, location-structured minimum escape actions over buses, controllers, cycles, and zones.
Basin boundaries	Saddles, periodic orbits, and stable manifolds organize attraction boundaries [7].	Use cycle geometry to seed and classify the critical boundary elements reached by the lowest-action physical disturbances.
Transient chaos	Chaotic saddles can produce complicated basin landscapes and spatially localized uncertainty [5].	Project branch-flow Lyapunov directions onto cycle space and test whether the same structures dominate controller feedback and escape.
Holonomy / matrix networks	Cycle holonomy constrains synchronization; quiver determinants admit cycle-holonomy expansions [8], [9]; matrix-weighted synchronization is an active theory [10].	Interpret $\det H_C$ as the full Abelian cycle invariant and noncommuting special-linear factors as controller-induced non-Abelian frustration.

D. Local cycle-flow boundary radius

Let $G_C = C^\top K_e^{-1} C$ and $c_e^\top = e_e^\top C$. The remaining flow slack on line e is

$$s_e = a_e - |\varphi_e^*|. \quad (11)$$

Consider the smallest cycle-coordinate displacement, in the G_C metric, that uses pure circulation to consume this slack.

Proposition 2 (Local cycle boundary radius). *The optimization*

$$\min_q q^\top G_C q \quad \text{s.t.} \quad c_e^\top q = s_e \quad (12)$$

has value

$$J_e^* = \frac{s_e^2}{c_e^\top G_C^{-1} c_e}. \quad (13)$$

Defining

$$\rho_{e,\text{cyc}} = \frac{s_e}{\sqrt{c_e^\top G_C^{-1} c_e}}, \quad \rho_{\text{cyc}} = \min_e \rho_{e,\text{cyc}}, \quad (14)$$

gives a local tangent cycle-flow feasibility radius.

Proof. The KKT condition gives $q^* = \lambda G_C^{-1} c_e / 2$. Enforcing the affine constraint determines λ , and substitution yields (13). $\square$

Remark 1. ρ_{cyc} is deliberately *not* called a transient-stability radius. It measures local cycle-flow distance to a sinusoidal feasibility boundary under a tangent metric. Its value in this work is as a geometric precursor and a hypothesis generator for the nonlinear escape computation in Section V.

IV. CONTROLLER-SHAPED FRUSTRATION

A. From phase frustration to matrix holonomy

For scalar phase shifts, the gauge-invariant information is the circulation

$$\Phi = C^\top \alpha \pmod{2\pi}, \quad (15)$$

which is the familiar topological frustration associated with nontrivial cycle phase. Recent twisted-Laplacian results show explicitly that nontrivial $U(1)$ holonomy can obstruct synchronization [8].

In a converter-rich network the port transport on an edge or reduced interconnection is instead matrix valued,

$$T_e(s) \in \text{GL}(d, \mathbb{C}), \quad (16)$$

and the cycle return map is

$$H_C(s) = \prod_{e \in C}^{\rightarrow} T_e(s). \quad (17)$$

Under local changes of port coordinates, H_C transforms by similarity. Thus its characteristic data are physical invariants of the closed loop.

B. Abelianization and the genuinely MIMO residual

For $d \geq 2$, the commutator subgroup of $\text{GL}(d, \mathbb{C})$ is $\text{SL}(d, \mathbb{C})$; equivalently, the determinant realizes the Abelian quotient $\text{GL}(d, \mathbb{C})_{\text{ab}} \simeq \mathbb{C}^\times$. Hence

$$\det H_C \quad (18)$$

contains the complete Abelianized cycle invariant. On a locally continuous branch of the logarithm, define

$$g_C(\omega) = \frac{1}{d} \log |\det H_C(j\omega)|, \quad (19)$$

$$\phi_C(\omega) = \frac{1}{d} \arg \det H_C(j\omega). \quad (20)$$

The scalar/isotropic limit $T_e = t_e I$ reduces ϕ_C to the sum of local phase lags around the cycle.

Choose locally a d th root $r_C = (\det H_C)^{1/d}$ and write

$$H_C = r_C \bar{H}_C, \quad \det \bar{H}_C = 1. \quad (21)$$

The factor $\bar{H}_C \in \text{SL}(d, \mathbb{C})$ carries the irreducibly multivariable part. For two cycles, a gauge-invariant obstruction to common scalarization is encoded spectrally by the group commutator

$$\mathcal{K}_{rs} = \bar{H}_r \bar{H}_s \bar{H}_r^{-1} \bar{H}_s^{-1}. \quad (22)$$

When $\mathcal{K}_{rs} = I$ for all cycle pairs and the holonomies are diagonalizable, the cycle family is Abelianizable in the sense developed in the companion scalarizability work. The present use is different: $\{g_C, \phi_C, \mathcal{K}_{rs}\}$ provides a frequency-dependent decomposition of controller-shaped cycle frustration that can be compared with nonlinear escape structures.

V. STRUCTURED MINIMUM TRANSIENT ESCAPE

A. Safe set and structured disturbance support

Consider a nonlinear semi-explicit DAE

$$\dot{x} = f(x, z, p) + B_S(x, z)u(t), \quad (23)$$

$$0 = g(x, z, p), \quad (24)$$

where S denotes the physical support of the disturbance. Depending on the study, S may be one bus, one line, one controller channel, a cycle-supported actuator set, or an aggregated zone. Define the safe operating set

$$\mathcal{S} = \{(x, z) : h_r(x, z) \leq 0, r = 1, \dots, n_h\} \cap \mathcal{B}(x^*), \quad (25)$$

where constraints can include angle cohesion, voltage/frequency ranges, current limits, PLL locking, dc-link bounds, and the basin of attraction $\mathcal{B}$ of the desired attractor.

Definition 1 (Structured transient escape action). For a disturbance support S , define

$$\mathcal{V}_S(T) = \inf_{u(\cdot)} \frac{1}{2} \int_0^T u^\top R_S u dt \quad (26)$$

subject to (23)–(24), $(x(0), z(0)) = (x^*, z^*)$, and $(x(t), z(t)) \in \partial \mathcal{S}$ for some $t \leq T$. The unrestricted-horizon action is $\mathcal{V}_S = \inf_{T>0} \mathcal{V}_S(T)$.

The most vulnerable physical support is

$$S^* = \arg \min_S \mathcal{V}_S. \quad (27)$$

This directly distinguishes *where the easiest perturbation enters* from where a mode has large participation.

B. Closed-form linear baseline

For the local approximation

$$\dot{\xi} = A\xi + B_S u \quad (28)$$

and an ellipsoidal safe boundary $\xi^\top Q \xi = 1$, define the weighted controllability Gramian

$$W_S(T) = \int_0^T e^{A\tau} B_S R_S^{-1} B_S^\top e^{A^\top \tau} d\tau. \quad (29)$$

Classical minimum-energy reachability gives

$$J_S^*(T) = \frac{1}{2\lambda_{\max}(Q^{1/2} W_S(T) Q^{1/2})}, \quad (30)$$

and the corresponding radius

$$\rho_S^{\text{lin}}(T) = \sqrt{2J_S^*(T)} = \frac{1}{\sqrt{\lambda_{\max}(Q^{1/2} W_S(T) Q^{1/2})}}. \quad (31)$$

Equation (31) is not new; it is included as the exact local benchmark that any nonlinear escape algorithm should recover in the small-disturbance limit.

C. Nonlinear optimality system

Ignoring DAE regularization details for notation, introduce an adjoint $\lambda(t)$ and Hamiltonian

$$\mathcal{H} = \lambda^\top f(x, z, p) + \lambda^\top B_S u + \frac{1}{2} u^\top R_S u. \quad (32)$$

An interior optimum satisfies

$$u^*(t) = -R_S^{-1} B_S^\top \lambda(t), \quad (33)$$

with adjoint dynamics $\dot{\lambda} = -\partial_x \mathcal{H}$ plus the algebraic multiplier equations associated with $g = 0$. The terminal transversality condition is supplied by the active safe-boundary constraint. Numerically, the linear solution from (31) provides a useful initial guess for direct collocation or multiple shooting; minimum-action/geometric methods provide an alternative when a small-noise quasipotential interpretation is desired.

D. Conservative barrier sensitivities

For a gradient-like lossless subsystem, let x^* be a stable critical point and $x^\ddagger$ an index-1 saddle governing an escape channel. Let

$$\Delta U = U(x^\ddagger, p) - U(x^*, p). \quad (34)$$

Proposition 3 (Barrier sensitivity). *If the stable point and saddle continue smoothly with parameter p , then*

$$\frac{d\Delta U}{dp} = U_p(x^\ddagger, p) - U_p(x^*, p). \quad (35)$$

Proof. Differentiate the two stationary values. Terms containing dx^*/dp and $dx^\ddagger/dp$ vanish because $\nabla_x U = 0$ at both critical points. $\square$

For an edge strength a_e and phase shift α_e in $U_e = -a_e \cos(\delta_e - \alpha_e)$,

$$\frac{d\Delta U}{da_e} = \cos(\delta_e^* - \alpha_e) - \cos(\delta_e^\ddagger - \alpha_e), \quad (36)$$

$$\frac{d\Delta U}{d\alpha_e} = p_e^* - p_e^\ddagger. \quad (37)$$

These identities give an immediate test of whether strengthening a line or shifting a controller-induced phase preference raises or lowers the conservative escape barrier.

Definition 2 (Transient-barrier reversal). A network modification exhibits a transient-barrier reversal over a specified model and safe set if a conventional small-signal/connectivity metric improves while the nonlinear escape action decreases, e.g.

$$\Delta\mu_2 > 0, \quad \Delta[-\max \Re\lambda] > 0, \quad \Delta\mathcal{V}^* < 0. \quad (38)$$

This definition is proposed as a falsification target, not reported as an achieved IEEE 39-bus result in the present draft.

VI. BASIN BOUNDARIES, CHAOS, AND CYCLE-RESOLVED STRETCHING

Recent work supports computing attraction boundaries through index-1 saddles, unstable periodic orbits, and their stable manifolds [7]. Separately, transient chaos in grid models can generate intricate basin structures and highly localized uncertainty [5]. The central question here is whether the same cycle structures that are soft in G_C , or active in the matrix feedback holonomies, also support the leading transient stretching near the lowest-action escape channel.

Let $\Phi(t, 0)$ be the tangent propagator along a nonlinear trajectory and let $v_1(t)$ denote a leading finite-time Lyapunov direction. A nodal angle perturbation alone is naturally a cut-space object, so a direct projection of $B^\top v_1$ onto cycle space is generally inappropriate. Instead construct the induced branch-flow perturbation

$$\delta\varphi(t) = D_x\varphi(x(t)) v_1(t). \quad (39)$$

With tangent compliance metric K_e^{-1} , define the weighted cycle projector

$$P_{\text{cyc}}^{K^{-1}} = C(C^\top K_e^{-1} C)^{-1} C^\top K_e^{-1}. \quad (40)$$

Then

$$\delta\varphi_{\text{cyc}} = P_{\text{cyc}}^{K^{-1}} \delta\varphi \quad (41)$$

quantifies the divergence-free component of the transient branch-flow stretching. A cycle-resolved chaos experiment should compare three supports: (i) the minimum-action path, (ii) $\delta\varphi_{\text{cyc}}$ along high-FTLE segments, and (iii) the dominant controller feedback cycles from $H_C(j\omega)$. Agreement would support a common mechanism; disagreement would be equally informative and would falsify an over-strong cycle-centric interpretation.

VII. FROM RED ZONES TO MINIMAL REMEDIATION

A single “weak-node score” would obscure distinct engineering questions. We therefore propose a multi-field stability atlas:

$m_{\text{lin}} = -\max_k \Re\lambda_k$	small-signal margin,
$\mathfrak{N}(\omega)$	scalar-model fidelity defect,
$(\chi_{\text{max}}, \rho_{\text{cyc}})$	cycle-geometric fragility,
$\mathcal{V}_S$	nonlinear structured escape action.

The first three can be screened cheaply relative to full nonlinear escape optimization. Only $\mathcal{V}_S$ answers the direct transient question.

A. Three different locations

For a critical case it is useful to distinguish:

- 1) *disturbance-entry support* $S^* = \arg \min_S \mathcal{V}_S$;
- 2) *failure structure*, e.g. the saddle/periodic-orbit manifold and cycle-flow region through which the trajectory exits;
- 3) *control hub*, the location where an admissible parameter change raises $\mathcal{V}^*$ most per unit cost.

There is no reason for these three locations to coincide. This distinction is an intended advantage over a single participation-factor or weak-bus ranking.

B. Least-cost dynamic remediation

Let κ collect controller gains, virtual damping/inertia, virtual impedance, BESS allocations, and optional network decisions. The intervention problem is

$$\min_{\Delta\kappa} C(\Delta\kappa) \quad (42)$$

$$\text{s.t. } \mathcal{V}^*(\kappa + \Delta\kappa) \geq \mathcal{V}_{\min}, \quad (43)$$

$$m_{\text{lin}}(\kappa + \Delta\kappa) \geq m_{\min}, \quad (44)$$

actuator, current, voltage, and bandwidth limits.

With an ℓ_1 or mixed-integer sparsity cost, the result can prefer a small controller change over a new BESS when the former provides adequate transient authority. This is the industrial endpoint of the framework: *retune first when possible; add damping/inertia/storage/topology only when required by the barrier constraint.*

VIII. PRELIMINARY IEEE 39-BUS CYCLE-GEOMETRY EVIDENCE

A. Setup

The preliminary experiment uses the solved MATPOWER IEEE 39-bus operating point and the same AC power-flow resolution procedure used in the companion MIMO scalarizability study. The original network has $n = 39$ buses and $m = 46$ active branches, hence cycle rank $\beta = 8$. The candidate line 25–38 is represented by $r = 0.0009$ p.u. and $x = 0.03$ p.u.; after inserting it the nonlinear AC power flow is resolved before computing the tangent quantities. For the geometric quantities in this section only, each branch is represented by the lossless synchronizing surrogate

$$a_e \approx \frac{V_i V_j}{x_e \tau_e}, \quad k_e = a_e \cos(\theta_i - \theta_j - \alpha_e), \quad (45)$$

including transformer tap ratio τ_e and shift α_e . This is a screening approximation; it is not the full SG/GFL/GFM transient model.

The same upgrade was previously found to increase standard Fiedler connectivity by 0.298% and generalized-Fiedler connectivity by 0.146% in the AC-consistent reduced study. Here we ask whether the cycle geometry gives the same qualitative assessment.

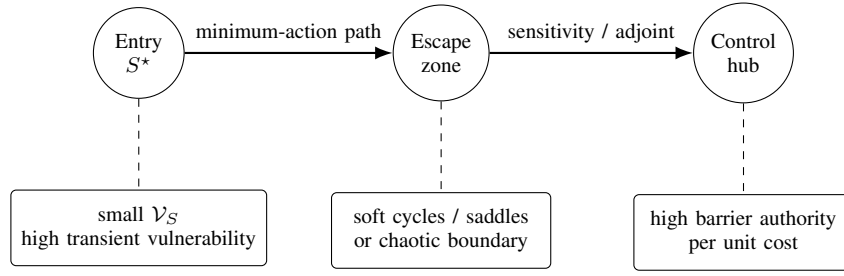
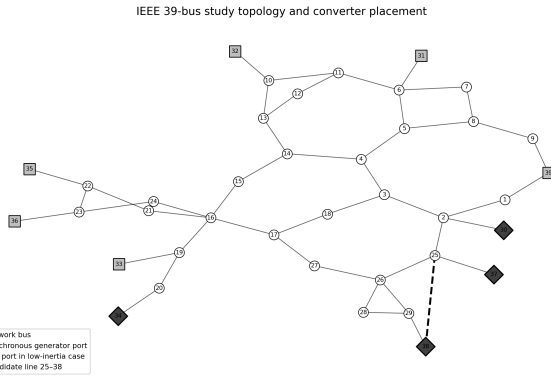
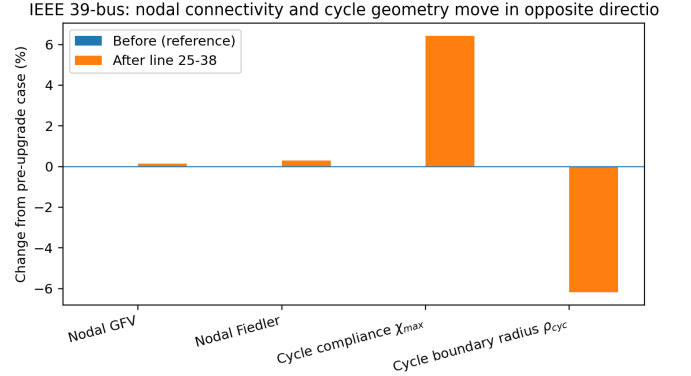


Fig. 2. The disturbance-entry point, the dynamic failure region, and the least-cost control location need not coincide. A principal objective of the proposed atlas is to separate these three “geographies” of transient instability.



(a) IEEE 39-bus topology and the 25–38 intervention.



(b) Nodal-connectivity versus cycle-geometry changes.

Fig. 3. Preliminary geometry experiment. The network upgrade is favorable under two nodal-connectivity measures but unfavorable under the two cycle-space screening quantities. This is a candidate geometric warning, not yet a demonstrated transient-stability reversal.

B. Cycle-compliance reversal

Before the upgrade, the maximum generalized cycle compliance is

$$\chi_{\max}^{\text{pre}} = 0.037848. \quad (46)$$

After adding 25–38 and resolving the AC operating point,

$$\chi_{\max}^{\text{post}} = 0.040277, \quad (47)$$

which is a

$$+6.42\% \quad (48)$$

increase in the softest cycle compliance. In parallel, the local cycle-flow boundary radius changes from

$$\rho_{\text{cyc}}^{\text{pre}} = 5.3085 \quad (49)$$

to

$$\rho_{\text{cyc}}^{\text{post}} = 4.9804, \quad (50)$$

a

$$-6.18\% \quad (51)$$

reduction. The limiting edge is 26–29 in both cases. Thus the same line addition gives

$$\Delta\mu_2 > 0, \quad \Delta\chi_{\max} > 0, \quad \Delta\rho_{\text{cyc}} < 0. \quad (52)$$

The nodal graph appears stronger, while the divergence-free cycle-flow geometry becomes softer and locally closer to its sinusoidal boundary.

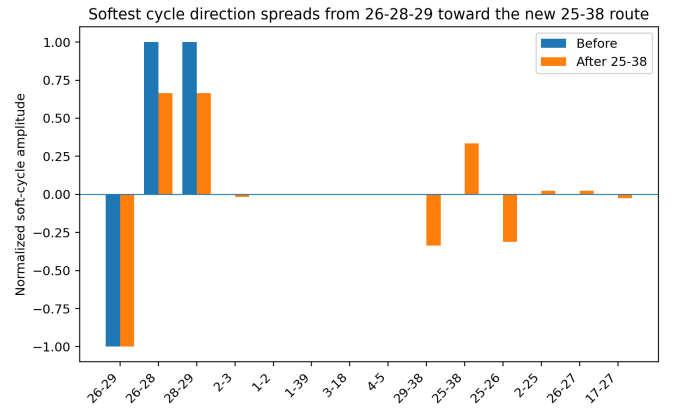


Fig. 4. Edge support of the softest cycle-compliance direction. Before the upgrade it is concentrated on the 26–28–29 triangle. After 25–38, the soft direction spreads toward 29–38, 25–38, and 25–26 while retaining the pre-existing triangular core.

C. Interpretation and limit of the result

The pre-upgrade soft direction is essentially the triangular circulation 26–28–29. After the upgrade, it retains that core while gaining substantial components along 29–38, the new 25–38 edge, and 25–26, as shown in Fig. 4. This supports a concrete hypothesis: a topologically beneficial intervention can connect a new route into a pre-existing compliant cycle zone.

However, (52) is *not* evidence that the nonlinear transient

barrier has fallen. The only valid conclusion at this stage is that two cycle-geometric screening objects move in the unfavorable direction while nodal connectivity improves. The next experiment must compute $\mathcal{V}_S$ and identify the actual basin-boundary element reached by the minimum-action path.

IX. FALSIFICATION-ORIENTED EXPERIMENTAL PROGRAM

The theory becomes publishable only if the following experiments distinguish it from existing ROA, CREP, and cycle-space analyses.

A. E1: Nonlinear minimum-action red-zone map

For each candidate bus, controller, line, and selected cycle zone S , solve (26) over a fixed family of safe-set definitions. Plot $1/\mathcal{V}_S$ on the network and compare against Fiedler/GFV, small-signal participation, CREP-like linearized rankings, $\chi_{\max}$ support, and ρ_{cyc} . The desired result is not high correlation; it is a physically interpretable ranking disagreement in which the nonlinear method identifies a low-action hidden red zone that conventional local metrics rank as benign.

B. E2: Triple ranking reversal

Search line additions, controller tunings, dispatches, and IBR placements for pairs A, B satisfying

$$\mu_{2,A} > \mu_{2,B}, \quad m_{\text{lin},A} > m_{\text{lin},B}, \quad \mathcal{V}_A^* < \mathcal{V}_B^*. \quad (53)$$

Finding (53) would demonstrate that better nodal connectivity and a better local eigenvalue margin can coexist with a smaller nonlinear escape barrier. Failure to find it over a broad physically admissible campaign would constrain the proposed claim and would itself be informative.

C. E3: Boundary element and cycle support

Use cycle-space equilibria to initialize a saddle search, then compute the stable manifold of the controlling index-1 element or unstable periodic orbit following modern DOA-boundary methods [7]. Compare the support of the boundary element with the soft cycle mode from (9). The cycle-geometry hypothesis survives only if this relationship is systematic rather than anecdotal.

D. E4: Transient chaos test

Compute finite-time Lyapunov exponents in a tube around the minimum-action path and use (41) to determine whether the leading stretching has a large cycle-flow component. Compare this support with the frequency-domain MIMO holonomies. Importantly, transient-chaos literature already reports relevant tree-like structures [5]; therefore the experiment is explicitly allowed to refute the hypothesis that cycles dominate every escape mechanism.

E. E5: Sparse remediation

After obtaining $\mathcal{V}^*$, differentiate the optimal value with respect to admissible controller parameters using the adjoint/envelope system and solve (42)–(44). Compare controller retuning, virtual damping, virtual inertia, BESS placement, and topology modification at equalized cost. A strong industrial result would be a case in which the cheapest barrier increase is achieved by a small remote controller change rather than storage at the failure location.

F. E6: Full-order validation

The decisive final gate is a full-order phasor/EMT replication with SG, GFL, and GFM models including current limits and inner loops. The reduced cycle-space objects should be computed from the same operating point, but transient conclusions must be validated in the full nonlinear model. Without this stage, the cycle geometry remains a principled screening theory rather than a complete transient-stability method.

X. INDUSTRIAL INTERPRETATION

The proposed method has three concrete industrial outputs.

A. A red-zone atlas with physical meaning

A red bus or zone is not red because an arbitrary index exceeds a threshold. It is red because the minimum admissible perturbation action $\mathcal{V}_S$ is small relative to the operating criterion. This makes thresholds directly interpretable in disturbance-energy, probability, or scenario terms.

B. Separation of cause and cure

The easiest disturbance may enter at one bus, the critical boundary manifold may be localized around another set of lines/cycles, and the most cost-effective control action may be remote. This is particularly relevant in converter-rich grids, where feedback closes through the network and local participation does not imply local controllability of the dangerous mechanism.

C. A hierarchy of interventions

The optimization in (42) can be configured to prefer low-cost retuning before hardware additions. The operational recommendation can therefore take the form: *retune two gains here; if the required barrier is still unmet, add virtual damping there; only then evaluate BESS or network reinforcement*. The methodology does not assume that more inertia, more damping, or a stronger line is universally beneficial; each action is evaluated by its effect on the structured escape barrier.

XI. LIMITATIONS

Four limitations are explicit. First, $\chi_{\max}$ and ρ_{cyc} are derived from a lossless tangent synchronization model; resistive and voltage-state extensions require a more general port metric. Second, the cycle rank can still be large in dense networks, so cycle-basis and reduced-subspace selection must be scalable. Third, non-gradient converter dynamics do not admit a unique physical energy landscape; the correct object is the action/quasipotential, not a forced spin-glass analogy. Fourth, the IEEE 39-bus results reported here establish only a *geometric* reversal. The minimum-action, saddle, chaos, and full-order EMT gates remain to be executed before claiming a nonlinear transient-stability reversal.

XII. CONCLUSION

This paper develops a research path from graph cycles to nonlinear transient escape without collapsing distinct concepts into one stability index. The cycle space supplies low-dimensional coordinates for circulation; the tangent synchronizing stiffness induces a basis-invariant cycle-compliance geometry; matrix-valued controller transport adds Abelian and non-Abelian cycle frustration; and structured minimum-action optimization defines the actual perturbation needed to leave a specified safe region. Preliminary IEEE 39-bus evidence shows that a connectivity-improving line can simultaneously soften the cycle geometry and reduce a local cycle-flow boundary radius. The key remaining question is whether that geometric warning predicts a smaller nonlinear escape action and the same physical exit channel in full-order dynamics. If so, the framework would support a practical workflow that identifies the weakest disturbance-entry zone, distinguishes it from the dynamic failure structure, and selects the minimum-cost controller or hardware intervention that raises the transient barrier.

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